

Data Centers Are Hotbeds of Health Risks | Article | EESI

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Advanced Search July 27, 2026 The United States has more than 5,500 data centers, with thousands more under construction or in planning stages. As more data centers are built, their impacts on neighboring communities are becoming increasingly evident. Data centers are associated with high energy and water use, higher utility bills for nearby households, and increased production of PFAS materials. EESI Data Center Resources See all of our articles on the impacts of data centers, including their energy needs, water usage, noise pollution, and contribution to higher energy bills: The Environmental Impacts of Data Centers (Article Series) All EESI Data Center Resources Data centers also affect public health. Impacts include carbon emissions from natural gas and coal power plants that are being built or kept online to power data centers. Data centers are also responsible for particulate matter emissions from on-site diesel generators that impair local air quality. It is reasonable to assume that fossil fuels will power new data centers, as 56% of their current energy already comes from natural gas and coal. Emissions from these power plants generate climate and health impacts throughout their lifecycle, including exposure to harmful pollutants and contamination of drinking water. Air pollution from fossil fuel plants in the U.S. costs each American an average of \$2,500 a year, totaling \$820 billion annually, and leads to more than 107,000 premature deaths. Data center developers are planning or constructing more than 90 gigawatts (GW) of on-site, off-grid natural-gas power plants to avoid long queues for grid interconnection. Off-grid gas turbines can be deployed modularly in less than two years, which is faster than traditional power plants, and can be scaled up to deliver power as needed. Unlike emergency diesel generators that run only during testing or emergencies, off-grid turbines operate 24/7, releasing air pollutants and greenhouse gases into the atmosphere and affecting nearby communities. By sidestepping the grid and utilities, data centers are increasingly securing reliable baseload power through behind-the-meter generation. Also called “speed to power,” it is the fast acquisition of power to keep up with exponential growth in AI demand and avoid potential disruptions from aging grid infrastructure and extreme weather. However, privately-owned gas turbines still pollute, although emissions might not be so easily measured. Communities located near a data center in Loudoun County, Virginia, have experienced health impacts from the facility’s natural gas turbines. According to a study prepared for the Piedmont Environmental Council, permitted particulate matter (PM2.5) emissions from these generators are estimated to result in between \$53–99 million per year in health-related damages, including premature deaths, school-loss days, emergency room asthma-related outcomes, and lost productivity. Due to data centers’ thirst for energy, coal is also making a comeback as some utilities are extending the lives of these plants by operating them longer than planned, even beyond their targeted retirement age. Running coal power plants past their

retirement lifetimes (in an uneconomically dispatched situation where operating costs exceed revenue) resulted in more than \$236 billion in health care costs in communities near these plants between 2015 and 2023. Throughout the coal power supply value stream, from mining to transportation and combustion, coal releases toxic gases, which impact public health. Coal power plants are blamed for 460,000 excess deaths between 1999 and 2020 due to the pollutants emitted by these plants. Coal power plants are the second largest emitter of carbon emissions and are a leading contributor to climate change. Air, water, and land pollution emitted by all types of fossil fuel power plants creates high health costs as pollutants lead to higher asthma rates and heart and lung disease. These health costs are in addition to the economic losses resulting from 900,000 lost workdays and over 3 billion lost school days nationwide between 2015 and 2023, as adults and children recover from the health impacts of emissions from fossil fuel plants. These health burdens translate to reduced quality of life and disproportionate impacts on vulnerable populations, such as children. U.S. health impacts specifically related to emissions from fossil fuel plants powering data centers accounted for \$6 billion in annual health costs in 2023, and by 2028 they are projected to reach \$20 billion annually due to increased hospitalization visits caused by health conditions, ranging from asthma to heart attacks. Diesel generators, the most common source of backup power for data centers, emit a cocktail of harmful pollutants, including particulate matter, carbon dioxide, nitrogen oxides, and sulfur oxides. Industrial-size diesel generators release 200 to 600 times more toxic nitrous oxides than a natural gas plant. These gases contribute to a host of health issues, including asthma attacks, worsening heart and lung disease, and even premature death. In general, air pollution disproportionately affects Black and Hispanic populations. As data centers continue to grow in size and number, so will the use of diesel generators. In some cases, diesel generators have been known to run beyond the permissible limits. During non-emergency situations, such as equipment testing, the Environmental Protection Agency (EPA) allows emergency generators to operate for up to 50 hours per year. If Northern Virginia data centers were to reach 100% of their permitted annual usage – in the case of an extended grid outage – the estimated annual public health cost would be \$2.2–3.0 billion. Additionally, data centers are not held to a federally-enforceable time limit when running diesel generators during grid failures, fires, and floods, provided the operator reports and logs the hours and the emergency. Noise Pollution from Diesel Generators On-site diesel generators are also a source of noise pollution from data centers. Noise levels from industrial-sized diesel generators can reach up to 105 decibels, as noisy as a jet flying overhead. In the 119th Congress, several bills have been introduced to address the environmental and health impacts of data centers. The Data Center Community Impact Act (H.R.7858) would require the Secretary of Energy to study the energy and health impacts of data centers on communities of color and low-income communities, particularly emissions from backup diesel generators and impacts on public health through pollution and heat island effects. The Artificial Intelligence Environmental Impacts Act of 2026, (H.R.9183 and S.4727) would require the EPA and other federal agencies to study the environmental and energy-related impacts of data centers. In addition, the Protecting Communities from Data Center Impacts Act of 2026 (H.R.9629) would require the National Academies of Sciences, Engineering, and Medicine to conduct an assessment of the environmental and public health impacts of data centers. On-site renewable-energy microgrids can provide reliable, clean energy to data centers, rather than fossil-fuel-based grid power or on-site natural-gas power plants. Zero-emission microgrids mitigate the adverse health impacts of powering data centers by reducing concentrations of air pollutants such as PM2.5 and nitrogen oxides emitted by on-site gas turbines. Battery storage enables microgrids to island themselves during a power outage and continue supplying critical power to data centers without noisy, polluting diesel generators. After a disaster, microgrids can connect to one another and to the overall grid, restoring power and frequency regulation to the energy grid. Microgrids on data center sites can also provide resilience to nearby communities by delivering continuous, clean power during grid outages, reducing peak demand, and serving as resilience hubs for community members. Proposed large data centers in Minnesota and

Oregon are already planning to add massive amounts of solar energy and battery storage to power their facilities. Author: Miguel Yañez-Barnuevo